Phase 4 Report

CMPT 276 – Introduction to Software Engineering

Group 8

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1. Game Overview

1.1 Game Description and Design Consistency

Our game is a turn-based, grid-based arcade-style game in which the player controls a main character navigating through a map to collect rewards and reach the exit. The core objective remains consistent with our original proposal: the player must collect all rewards and successfully evacuate the map to complete a level.

The gameplay is structured around a turn-based loop, where each player action is followed by enemy movements. As the player progresses through levels, the difficulty increases through changes in map complexity, as well as the number and behavior of enemies, traps, and rewards. This core loop and progression system were implemented successfully and remain faithful to the initial design.

In terms of implementation, all of core gameplay mechanics have been successfully developed and integrated until the last minute of the project.

1.2 Trap Mechanism

One of the key features in the game is the trap mechanism, which is a customized idea conducted as a unique gameplay element. When the player first encounters a trap, it can be collected without any penalty. The player can then deploy it strategically along their path using a control input (the space bar) to eliminate pursuing enemies.

However, once deployed, the trap becomes a risk as well as a tool. If an enemy steps on it, the enemy is eliminated. In contrast, if the player encounters the same trap again, the player will be eliminated instead. This creates a balanced risk–reward system that encourages careful planning and strategic use.

This mechanism enhances gameplay by allowing players to actively respond to enemies rather than relying solely on avoidance.

1.3 Changes from the Original Design

The most significant changes from the original plan lies in the movement behavior of the moving enemies. Originally, we aimed to create a fast-paced and highly challenging gameplay experience. To achieve this, enemies were designed to closely follow the path of the main character at the same speed, creating a constant and intense sense of pressure on the player.

However, based on feedback of TA after Phase 2, particularly the concern that the enemy behavior was overly harsh for players that left little room for recovery after mistakes, the chasing algorithm was redesigned. The updated implementation reduces the aggressiveness of enemies by slowing their movement and making their behavior more forgiving. This adjustment improves accessibility and makes the game more approachable for a wider range of players.

Nevertheless, this change introduces a trade-off. While the game becomes more balanced and less frustrating for general players, it may reduce the level of excitement for players who prefer a more intense and high-pressure experience.

1.4 Lessons Learned

An important lesson learned from this project relates to user interface (UI) design. Unlike other aspects of the system, most team members had little to no prior experience in designing and implementing UI components. As a result, a significant amount of time was spent exploring design approaches, experimenting with layouts, and refining visual elements to achieve a coherent and usable interface.

This process revealed that UI design is about usability, clarity, and how effectively information is communicated to the player. Decisions such as layout organization, visual hierarchy, and element visibility required multiple iterations and careful consideration, especially when integrating with gameplay logic.

Although the final interface may appear relatively simple compared to other projects, it represents a substantial learning experience.

2. Game Tutorial

2.1 Initial Game Screen

The first appearance when installing the game

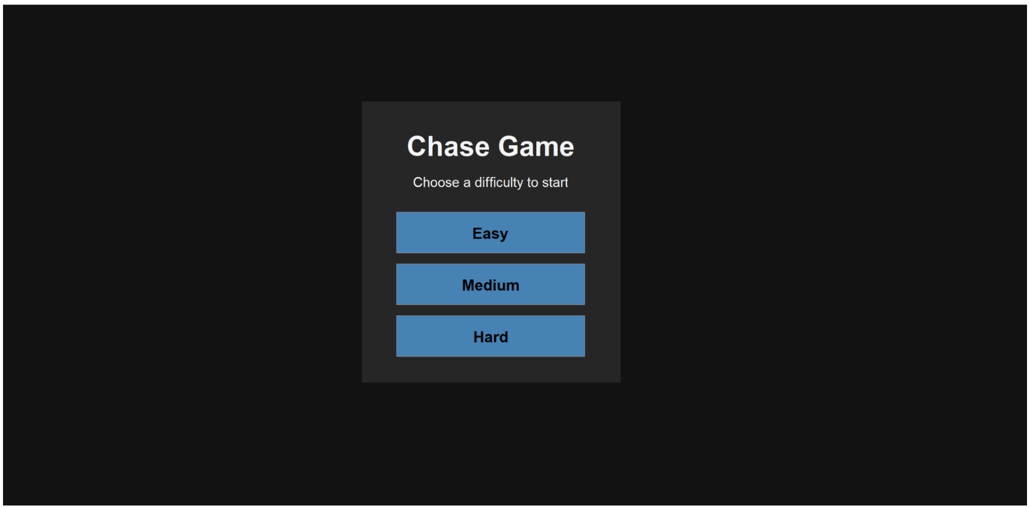

Figure 1: Initial game screen with difficulty selection and basic instructions

When the game launches, the player sees an initial screen that serves as the entry point. This provides the user with a choice of difficulty and a basic guidance on how to start and ensures a clear and simple first-time user experience.

2.2 Instruction and Confirmation Interfaces

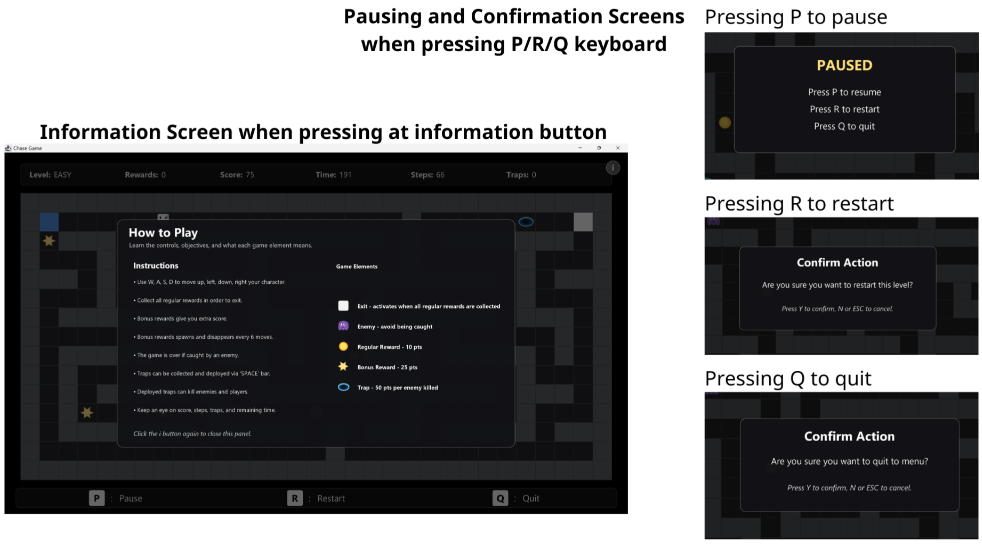

Figure 2: Instruction panel and confirmation dialogs

2.3 Main Interface and Controls

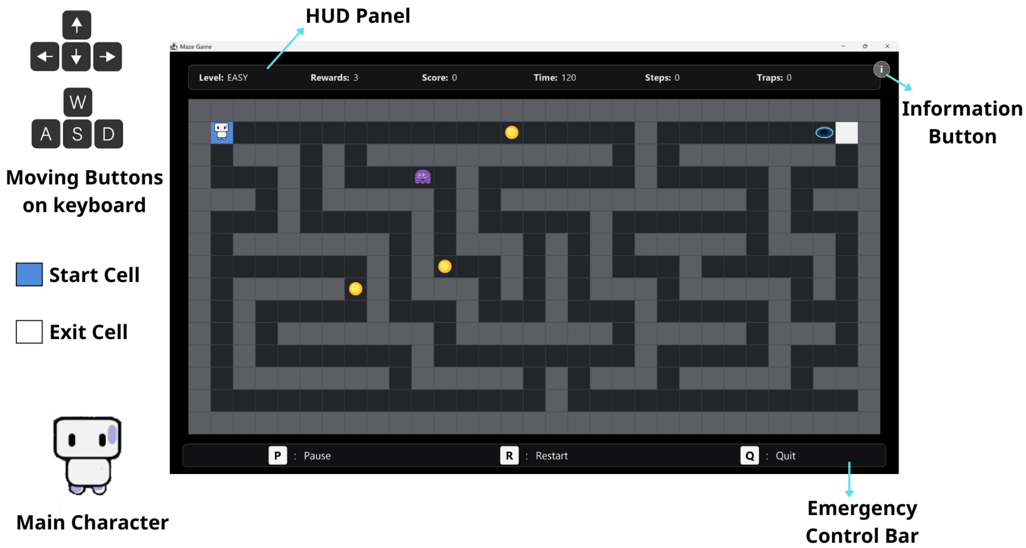

Figure 3: Main game interface with HUD and control elements

The main interface represents the main game and displays key information such as the Level difficulty, the rewards left to consume, the user's current score, his current number of steps, the number of traps and the time left to complete the level.

2.4 Trap Mechanism Demonstration

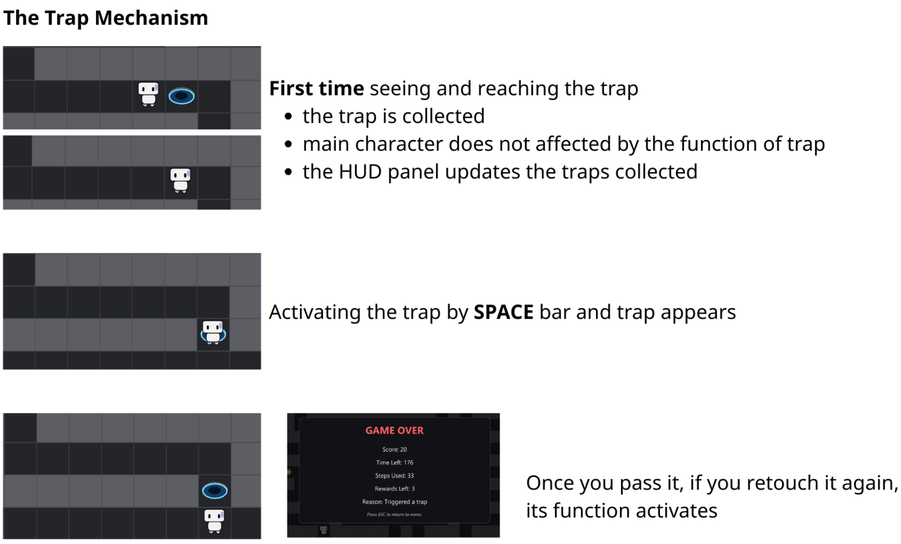

Figure 4: Demonstration of trap collection and deployment

Traps can be collected and later deployed to eliminate enemies. However, once placed, they become dangerous to both enemies and the player. This creates a risk-reward system that requires careful planning.

2.5 Gameplay Flow and Outcomes

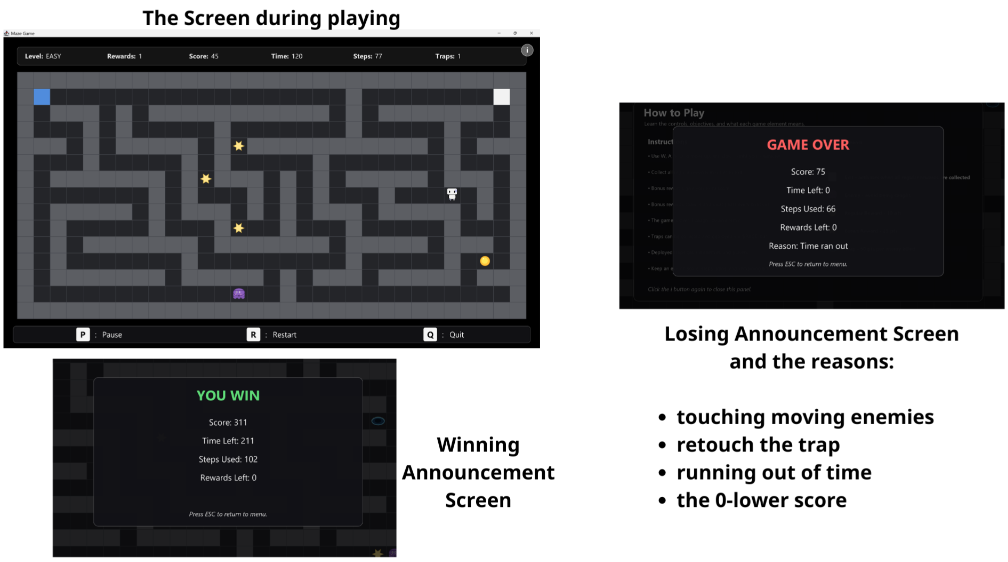

Figure 5: Gameplay screen with win and loss outcomes

The player moves through the map to collect all rewards while avoiding enemies. The game follows a turn-based system where enemies move after player action. The player wins by collecting all rewards and reach the exit, and loses if caught by enemies or traps.

Whether the user completes the game or fails to do so, an interface showing the results of the game is shown after the level is done.